



ZÜRICH UNIVERSITY OF APPLIED SCIENCES
DEPARTMENT LIFE SCIENCE AND FACILITY MANAGEMENT
INSTITUTE FOR COMPUTATIONAL LIFE SCIENCES

Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm

The Influence of Cold Water Tributaries on Cold Water Patches

Bachelor Thesis

by
Dürig Etienne

Bachelor program Applied digital Life Sciences
Focus Digital Environment
Study cohort 2023

June 14, 2026

Dr. Manuel Antonetti

ZHAW Life Sciences und Facility Management
Institute for the Environment and Natural Resources
Research Group: Ecohydrology
Campus Grüenthal, 8820 Wädenswil

Dr. Norman Juchler

ZHAW Life Sciences und Facility Management
Institute for Computational Life Sciences
Research Group: Medical Image Analysis and Data Modeling
Schloss, 8820 Wädenswil

Contents

1	Introduction	3
2	Theoretical Background	5
2.1	Model - Working Definition	5
2.2	Model Input Sensitivity Analysis	5
2.3	Sensitivity Analysis	5
2.3.1	The Morris Method	6
2.4	Coldwater Patches	9
2.5	Data	9
3	Methods	10
3.1	Optimizationn of Existing Algorithm	10
3.1.1	Reimplementation of the CWP algorithm based on GDAL and sys- tem calls	11
3.1.2	Output equivalence and Runtime Comparison	12
3.2	Sensitivity Analysis using Morris	13
3.2.1	Determining and classifying step length sensitive CWP	13
3.2.2	Computing morris sensitivity indices with respect to CWP area	16
3.3	Usage of KI	18
4	Results	19
4.1	Comparison of outputs	19
4.2	Speed comparison in execution	20
4.3	Sensitiviy scores for step length parameter	21
4.4	Mu star for all parameters over all CWP	22
4.5	Mu.star and size for every cwp	22
5	Discussion	24
5.1	Difference in cwp polygons between old and new function	24

5.2 Speed comparison 24

5.3 Answering research question 1 25

5.4 Overall Sensitivity to the different Paramters 26

6 Optimisation of the CWP algorithm based on the morris sensitivity analysis 27

6.1 Deploying insight from the morris screening 27

6.2 Operationalisation of Surrounding Stream temperature 27

6.3 Implementation Detail 28

6.4 Reduction of parameters 28

6.5 Quellenverzeichnis 29

Chapter 1

Introduction

Thermal heterogeneity in river systems is a vital component of aquatic biodiversity. Specifically, the presence of cold-water refugia is essential for the survival of thermally sensitive fish species, as river temperatures in Switzerland are expected to rise due to climate change. Currently there is no comprehensive and extensive information about the presence, scale and location of such cold-water patches in water bodies available in Switzerland. This data, however, is critical for river management in the context of climate change ([Tonolla and Antonetti, 2021, 2026](#))

As part of a cantonal project aiming to quantify the temperature heterogeneity and detect cold-water patches in the Emme River and its tributaries, TIR imagery covering over a 100 km river transect where collected in a helicopter-based survey. These TIR images where then processes to extract longitudinal and transversal temperature gradients, as well as to detect cold-water spots. The current processing workflow involves stitching highly overlapping TIR images into orthophotos and performing thermal corrections using in-situ temperature loggers. Then, longitudinal and transversal temperature gradients are derived. Additionally, a newly developed algorithm for the automatic detection of cold-water patches is deployed with the goal of mapping these patches.

The current method for the automated detection of cold-water patches (CWPs) relies on four externally configured hyperparameters: step length, buffer width, temperature delta , and minimum area. Sensitivity analyses from previous studies indicate that the detection results vary significantly based on the chosen parameter values, with step length and having a particularly strong influence on the variance of the algorithm’s output. The primary aim of this thesis is to systematically investigate the algorithm’s variance with respect to its hyperparameters, specifically focusing on how tributary inflows and adjustments to the processing workflow changes interact with the variance caused by specific hyperparameters.

The algorithm defines a CWP by comparing individual pixel temperatures against a local median reference temperature calculated over a defined segment (step length). It is hypothesised that the choice of step length introduces significant variance near the inflow of tributaries. This, because cold- or warm-water tributaries introduce sharp longitudinal temperature gradients which can skew the calculation if the step size is chosen sub-optimally

Research question 1: How does the step length contribute to the algorithms variance in river sections containing a tributary with significantly differing water temperatures, compared to sections without such tributaries?

H0: The variance introduced by the step length parameter is not influenced by the presence of a tributary with a substantially different temperature.

H1: The variance introduced by the step length parameter is significantly influenced (increased or decreased) by the presence of a tributary with a substantially different temperature.

Research question 2: What adjustments could be made to the detection-algorithm to reduce variability with respect to parameters while also achieving plausible detection results?

Chapter 2

Theoretical Background

2.1 Model - Working Definition

The term model is widely used across environmental and earth sciences, yet a precise definition of a model is surprisingly challenging to find in literature. In this work, a model is understood as a set of simplified laws and processes, intertwined by dependencies, that together form a reduced reflection of an underlying system. The aim is to capture the essential behavior of that system and enable analysis from which inferences about reality can be drawn. This view is consistent with (Saltelli et al., 2007), who, drawing on Rosen (1991), describe the modelled portion of reality as an arbitrary enclosure of an otherwise open and interconnected system.

2.2 Model Input Sensitivity Analysis

The inputs to models can diver and are generally dependent in their structure and many-foldness on the nature of the model itself. Under the light of sensitivity analysis however, it can be worthwhile to define, what the inputs to a model actually contains. In sensitivity analysis, everything that causes variation in the output of the model is classified as an input (Saltelli et al., 2007). Providing an example. If a linear regression of a target variable Y is performed, then not only are the predictors $X_1, \dots, X_n$ considered as model inputs, but also the parameters $\theta_1, \dots, \theta_n$ that correspond to the predictors.

2.3 Sensitivity Analysis

According to Saltelli et al. sensitivity analysis is “The study of how uncertainty in the output of a model [...] can be apportioned to different sources of uncertainty in the model input” (Saltelli, 2004). Sensitivity analysis provides a range of benefits. Firstly it allows the Modeler to apply a strategy called “Factor Fixing”. If we discover, that the model output Y is independent of one or multiple parameters θ_i , then we can set these to fixed constants. It may then even be possible to condense a complete submodule of a model, if all parameters connected to this sub-module are noninfluential. So sensitivity analysis can be a tool for model simplification (Saltelli et al., 2007). Secondly, sensitivity analysis generally improves model validity by identifying highly influential and therefore critical

model components (Ligmann-Zielinska, 2013). A range of different methods for sensitivity analysis exist. They can be grouped into global and local analysis methods.

Local methods compute a sensitivity score for each parameter at a single point in the input space, typically the baseline point, which is customarily the best estimate point for the parameters. Scores are based on partial derivatives and are therefore only informative for linear models, where the derivative at one point represents behaviour across the entire input space (Saltelli and Annoni, 2010).

Global sensitivity analysis methods explore the entire input space by varying all input parameters simultaneously across their plausible ranges. Sensitivity scores are therefore not tied to a single point but reflect average behaviour across the full parameter space. This makes global methods valid regardless of model linearity, as no assumption about model structure is required (Saltelli and Annoni, 2010).

global methods

2.3.1 The Morris Method

The Morris Method belongs to the group of screening methods in sensitivity analysis. These are ideal to filter a short list of important input parameters from computationally expensive models, with a great number of input parameters. This can be done, as these methods only require a relatively little number of model evaluations to provide sensitivity measures for the input parameters. This makes these methods a good choice when the target of the sensitivity analysis is Factor Fixing. (Saltelli et al., 2007). The following explanation of morris method closely follows (Saltelli et al., 2007) and is explained more explicitly in their ...

At the core of this introduction sits a k -dimensional model Y with inputs $X_1, \dots, X_k$. These inputs all together can also be written in a vector $\mathbf{X}$. The model inputs are defined in the k -dimensional unit cube, meaning that X_i can take on a value in the interval $[0, 1]$ (Saltelli, Tarantola). It is worth pointing out, that every model with a continuous input can be scaled to the $[0, 1]$ range, which makes the unit cube a great place to define methods (Claude). In the Morris method, this input space is discretized into a grid Ω . This grid features p levels.

An elementary effect of the i th input variable is then defined as:

$$EE_i = \frac{Y(X_1, X_2, \dots, X_{i-1}, X_i + \Delta, \dots, X_k) - Y(X_1, X_2, \dots, X_k)}{\Delta} \quad (3-1)$$

Where Δ represents some step in the unit cube. Δ can be chosen from a range of different values, which are given by the set $\left\{\frac{1}{p-1}, \dots, 1 - \frac{1}{p-1}\right\}$. Which value to choose for Δ from this set has implications and will be addressed in a later paragraph.

In the definition of the elementary effect, the vector $\mathbf{X} = (X_1, X_2, \dots, X_k)$ has to fulfil the condition that $\mathbf{X} + \mathbf{e}_i \cdot \Delta$ still has to lie within the grid Ω . $\mathbf{e}_i$ is a vector of zeros except having a one at its i th entry. So for instance:

$$\mathbf{e}_2 = (0, 1, 0, \dots, 0) \quad (3-2)$$

The Morris method occupies itself with finding the statistical distribution of each of the i elementary effects. $\mathcal{F}_i$ is thereby used to denote the distribution of EE_i , in other words

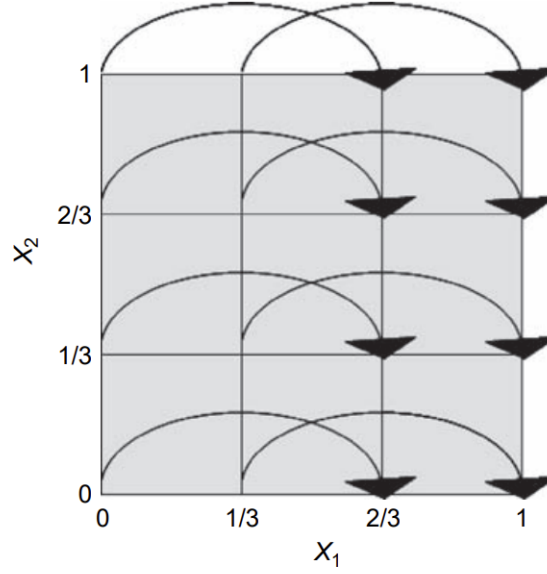


Abb. 3-1: An example grid Ω . The grid features two parameters X_1, X_2 , meaning that it is a two-dimensional grid. Further, it features four levels ($p = 4$). The step length Δ chosen here is $\frac{2}{3}$, which corresponds to an element of the set of viable values for Δ : $1 - \frac{1}{4-1} = 1 - \frac{1}{3} = \frac{2}{3}$.

$\mathcal{F}_i \sim EE_i$. $\mathcal{F}_i$ is a finite discrete distribution, and its number of elements depends on the number of grid levels p , which is however not that much of interest in an applied setting.

How are the EE_i sampled? Generally many different points $\mathbf{X}$ over the whole grid are sampled and then used to compute elementary effects EE_i . This guarantees, that each elementary effect EE_i is computed for many areas within the grid. This is also the reason, why the Morris method is generally classified as a global method. The exact sampling scheme is out of this project's scope but is covered in detail in Saltelli_Marco chapter 3.3.

Once the distributions $\mathcal{F}_i$ have been mapped out, statistical measures can be computed just like over every other statistical distribution. In Morris, the mean μ_i and the standard deviation σ_i of $\mathcal{F}_i$ are computed. These two statistical measures together already allow for reasonable insight into the influence of the i th parameter X_i which corresponds to the i th elementary effect EE_i :

If μ_i deviates strongly from 0, then this means that the elementary effect EE_i generally takes on non-zero values and therefore that the model output changes notably when a step of Δ is performed along the dimension spanned by X_i . This then ultimately means, that changes in X_i seem to generally affect the model output Y considerably and X_i therefore receives a high sensitivity score. Generally, the more μ_i deviates from zero, the larger the sensitivity of the model output towards X_i .

If σ_i is large, this means that the distribution $\mathcal{F}_i$ is spread out considerably over a larger range. This indicates, that the elementary effects EE_i that were computed were generally different depending on their location in the unit cube. Meaning that in one area, the model might only yield small changes in output, when X_i is varied while in other areas X_i leads to strong changes in model output. This behaviour points either to nonlinearity regarding X_i or to the fact that X_i interacts with other input parameters. Which effect is responsible for the high standard deviation (or whether both are responsible for the

effect at the same time) can not be determined. Which is a shortcoming of the method of Morris. Figure 3-2 provides an visual example of possible F distributions of a Factor X_i with the associated μ and σ values and the corresponding interpretation.

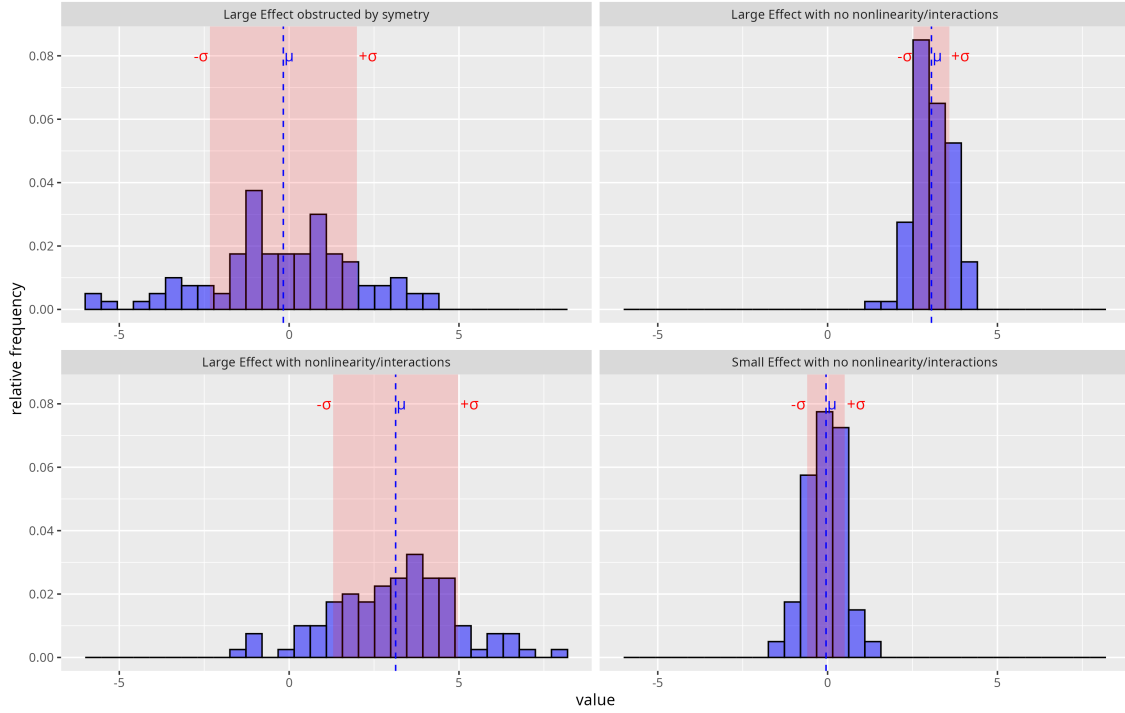


Abb. 3-2: Different F distributions of a parameter X_i are shown. Additionally, μ and σ of each distribution is provided. An interpretation is provided

Situations can occur, where $\mathcal{F}_i$ contains many extreme values however the distribution is symmetric around zero. Then computing the mean value μ_i will yield a value close to zero and we would assume, that the parameter X_i therefore is noninfluential. This would be a faulty assumption, as the sampled elementary effects EE_i are substantial in their magnitude but just cancel themselves out. This phenomenon can be spotted by always taking σ_i into consideration, as in this scenario σ_i is large, but it is more common to also sample an additional distribution $\mathcal{G}_i$ which captures the magnitude of the elementary effects $|EE_i|$. So $\mathcal{G}_i \sim |EE_i|$. We then compute the mean of this distribution μ_i^* which can be interpreted like μ_i . The further away from zero, the more influential is the parameter. But since it is the magnitude, the distribution never features negative values and therefore cancellation effects can not occur.

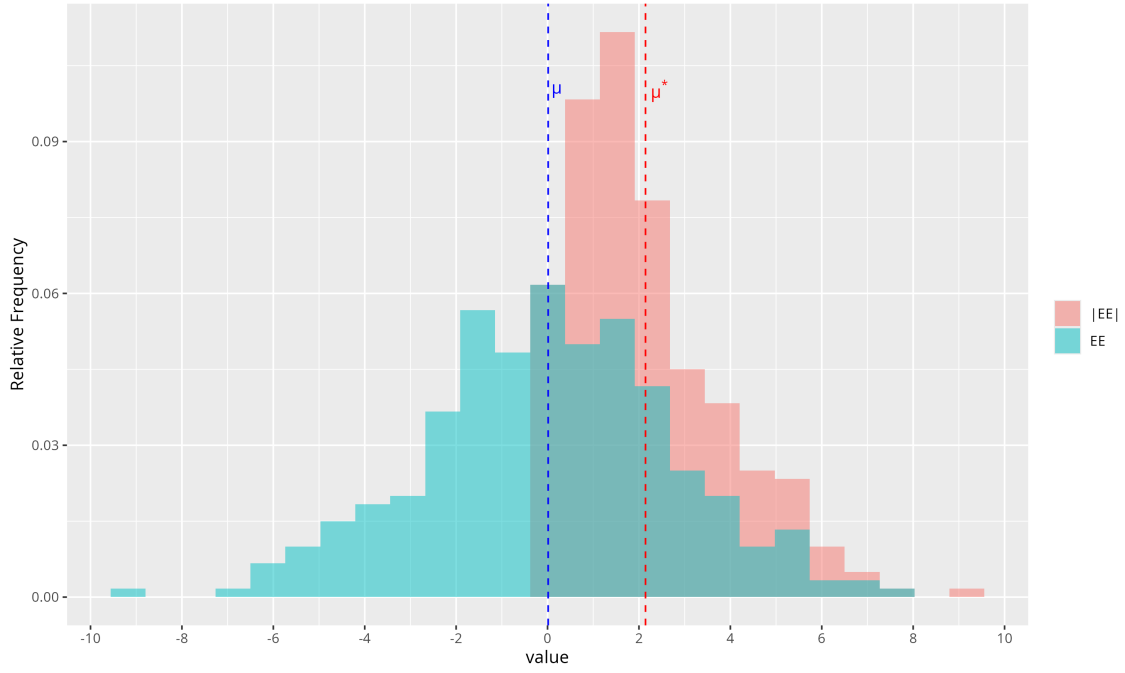


Abb. 3-3: The difference of μ and c visualized with their underlying distribution $\mathcal{G}$ and $\mathcal{F}$. While the mean μ of the distribution $\mathcal{F}$ lies at around zero, μ^* significantly deviates from zero

2.4 Coldwater Patches

Cold water patches are areas in rivers and streams that feature a colder water temperature than the temperature of the "hauptgerinne". The temperature difference varies in literature but is typically set to be between 0.5 and 3 degrees Celsius below the "hauptgerinne" temperature, but this varies over the literature (z.B. Dugdale et al., 2013, 2015; Wawrzyniak et al., 2016; Casas-Mulet et al., 2020). Cold water patches are not automatically a cold water refugia. These are coldwater patches which are also used by coldwater organisms for thermal relief (Meja et al.). The importance of cold water refugia is expected to rise with the effects of climate change (IPCC, 2014)

2.5 Data

Chapter 3

Methods

The presented methods can be divided into 3 logical steps. The first step focuses on optimizing the currently available CWP detection algorithm with respect to execution speed, to set the stage for the sensitivity analysis. The second step focuses on deriving sensitivity indices using the morris method for a set of variable CWPs which can be clearly attributed to being tributary-caused or non tributary-caused. The third step of the method then focuses again on adjusting the CWP algorithm, this time with respect to the insight generated from observing the sensitivity to different parameters. This three step division of the methods is also provided as a graphical overview in Fig 0-1.



Abb. 0-1: High-level overview of the methodological workflow.

The first optimisation step and the second step of the methods was performed based on the data collected of the emme unterlauf. The sensitivity analysis is performed using all the data from both segments of the emme unterlauf aswell the segment of the emme oberlauf.

3.1 Optimizationn of Existing Algorithm

most sensitivity analysis methods feature a high cost when it comes to model executions. Morris, is one of the cheaper methods but still requires a reasonable amount of model executions. A model with 3 parameters and $r=50$ requires $50 \cdot (3 + 1) = 200$ executions. To perform these executions in a feasonable timespan for the provided CWP algorithm optimisation of the computations is needed. Additionally parallel execution of the algorithm should be performed to further reduce execution times for the sampling. For the latter the zhaws High perfomance cluster is beeing used which has direct implications on

the optimisation approach. To ensure that the code optimisation speed gains also carry over to the infrastructure (e.g. the underlying distributed file system) of the hpc, the optimisation and benchmarking is performed directly on the hpc system.

3.1.1 Reimplementation of the CWP algorithm based on GDAL and system calls

The current algorithm focuses on per slab processing of the raster using a set of R loops and relying on `terra` and `sf` functions to implement the logic. The reworked algorithm is disk/file based and deploys GDAL over its command line interface directly. This decision was chosen, as `terra` operations are sometimes not too performant and GDAL offers a often faster set of geocomputational tools which can be called from R using `system()` or `sf::gdal_utils()` (Lovelace et al., 2019). Further, empirical testing in a prior coursework exercise also showed a significant speedup of GDAL-based raster operations compared to `terra` across a range of use cases (Dürig, 2026). This coursework was not peer-reviewed and is cited here only as a supporting empirical observation. Additionally the GDAL approach was also chosen, as it enables easy chunk processing with reading and writing to disk, which helps to keep memory footprint small. This can be beneficial in a HPC setting, where not only the number of available cores but also the available memory can be come an issue in performing computations. The drawback of this decision is that writing intermediate results to disk causes large spikes in disk space consumption, as intermediate rasters of the algorithm can be up to 25 GB large. This was counteracted by deploying compression during writing and decompression during reading, a strategy that only produced little computational overhead but greatly reduces the size of the intermediate results. Specifically, this can be implemented by setting the compression options `COMPRESS=DEFLATE` and `PREDICTOR=2` when working with GDALs geotiff driver (GDAL/OGR contributors, 2026). Lastly it should also be pointed out that heavily relying on `gdal` introduces a lot of I/O activity, which can significantly hurt performance on distributed file systems. To further optimize execution speed, `terra::extract()` was replaced with `exact_extract()` from the `exactextractr` package. This was done as `exact_extract()` extracts aggregated values from rasters considerably faster than `terra::extract()` for large rasters (Mieno, 2022); it should be noted, however, that this speed difference was not tested in isolation within this work. A further difference between the two functions concerns how partially covered cells are handled: `exact_extract()` performs a coverage-fraction-weighted extraction, in which partially covered cells are weighted by their fraction of coverage (ISciences, LLC, 2026), whereas `terra::extract()` by default includes a cell only if its center point falls within the polygon (Hijmans et al., 2024) – the behaviour used in the original (v1) implementation. To approximate this cell-center-inclusion behaviour within `exact_extract()`, only cells with a coverage fraction of at least 50% were included. This approximation is not exactly equivalent to `terra`’s center-point rule.

Table 1-1 provided a high level overview over the reimplemented cwp detection algorithm. The Description columns lists the computational steps of the algorithm, while the Tool / Function lists the main important function used in this step.

Tab. 1-1: High-level overview of the reworked CWP detection algorithm, including the primary tool or function used for each step.

Step	Description	Tool / Function
1	Divide the river reach into n centerline segments; construct a wide zone polygon and a narrow reference-strip polygon for each segment.	<code>sf::st_buffer</code>
2	Rasterize the zone polygons to a raster matching the TIR raster resolution, using the segment id as burn-in value.	<code>gdal vector</code> <code>rasterize</code>
3	Round the TIR raster using a fixed rounding factor.	<code>gdal raster calc</code>
4	Extract the median temperature per segment from the rounded TIR raster within each reference strip; store in a look-up table keyed by segment id.	<code>exactextractr::exact_extract</code>
5	Reclassify the zone raster, replacing each segment id with its reference temperature from the look-up table, yielding the reference raster.	<code>gdal raster</code> <code>reclassify</code>
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.	<code>gdal raster calc</code>
7	Polygonize the binary CWP raster into patch polygons.	<code>terra::as.polygons</code>
8	Filter patches by minimum area, compute per-patch temperature statistics, and apply a patch-level threshold check against the reference temperature.	<code>exactextractr::exact_extract</code>

3.1.2 Output equivalence and Runtime Comparison

The resulting reimplementation was tested for equivalence in output by comparing the produced CWP polygons to the CWP polygons of the old algorithm when deployed with the exact same set of parameters. Table 1-2 shows the parameter combinations deployed in testing. Further, the runtime between the new algorithm and the existing version was compared with respect to the step length parameter. Runtime was compared across three segment lengths (400, 800, and 1200 m), with all other parameters held constant (zone halfwidth = 60 m, buffer = 3 px, $\Delta C = 1.0^\circ\text{C}$, minimum patch area = 2 m^2 , rounding = 0.1°C , 8-connectivity enabled).

Tab. 1-2: Parameter combinations used for the output equivalence comparison between v1 and v2.

Parameter	Test 1	Test 2	Test 3
segment_length_m	200	800	1200
zone_halfwidth_m	60	60	60
buffer_px	4	3	5
delta_C	0.5	1.0	1.5
min_patch_area_m2	2	3	2
round_to	0.1	0.1	0.1
connect_diagonals	TRUE	TRUE	TRUE

3.2 Sensitivity Analysis using Morris

With the algorithm optimized, sensitivity analysis can now be carried out to determine the sensitivity of the CWP detection algorithm with respect to the parameters buffer pixel count, step length and temperature delta. Especially of interest is to observe the difference in sensitivity towards the step length parameter between tributary-caused and non tributary-caused CWP. For this reason a pre selection method is deployed to guarantee that the studied CWP are sensitive towards the step length parameter. Sensitivity towards pixel buffer count and temperature delta are also of interest. (Tonolla et al.) already found in their sensitivity analysis, that the buffer pixel count seems to have little influence on the CWP area. If this finding can be replicated algorithmic simplification using factor fixing (reference to theory) might be possible. Additionally, the temperature delta parameter is among the most influential according to Tonolla et. al. It is of general interest a tool to determine the validity of the method, to check whether the same is found. The general approach can be logically segmented into two steps, section 1 focuses on determining CWP that are susceptible to the step length parameter and classifying these cwp into tributary-caused and non tributary-caused CWP. The second step focuses on deriving morris sensitivity indices for these CWP with respect to their area.

3.2.1 Determining and classifying step length sensitive CWP

Figure 2-2 shows a high level overview of the first step. First a morris design with $r=44$ and $k=2$ is step up for the buffer pixel count and step length parameters. Then the speed optimized CWP algorithm is deployed on the resulting parameter tuples. Lastly the results are aggregated to a fuzzy representation of the CWP over all runs, to then select CWP which are variable to the CWP algorithm and to classify them into tributary-caused and non-tributary caused CWP.

**Abb. 2-2:** TODO: describe what this overview plot shows.

Setting up first Morris experiment

First, a two dimensional morris sampling grid was set up spanning two parameters buffer pixel and step length. A total of 6 levels were set for each parameter. the step length varying from 400 m to 1200 m and the number of pixels in the buffer spanning from 2 to 7. To compute the partial differences a jump distance of 3 was set. Fig shows the deployed grid visualized in greater detail with the jump distance additionally drawn in.

From this grid a total of 44 trajectories were sampled ($r=44$) which results in a total of $66 * (2 + 1) = 132$ parameter tuples being generated. Each tuple holds a different set of buffer pixel count and step length. These were written to a file along with a unique identifier of the tuple. For the sampling procedure the R package "sensitivity" was deployed. This package additionally produces an object representing the morris sampling grid and can later be provided with the results to carry out the computation of the morris indices automatically. this object was saved to an RDS and stored on disk for later computation.

Model Evaluation along Parameter Combinations

Next a parallelized computation scheme was set up to evaluate the CWP algorithm along the created parameter tuples while keeping the other parameters not part of the morris experiment constant. Table x shows to which values the other parameters were held constant.

To orchestrate the parallel execution of the CWP algorithm, the open source software GNU parallel was deployed (Tange, 2024). Specifically gnu parallel was used to implement a worker pool of processes that each execute the cwp algorithm with a given parameter tuple they fetch from a parameter text file. Before starting the computation, the processes also set up an individual working directory where intermediate results as well as the final result are written to. To keep the parameter combination as well as the id present and the results therefore identifiable, the folder name holds the id as well as the parameters.

Figure 2-3 visualizes the GNU parallel workflow deployed as an execution graph in time where time is encoded in the y-dimension of the plot.

Creation of a fuzzy overlay of CWP

the resulting cwp polygon layers were then processed into an singular raster over a series of processing steps. First each polygon layer was rasterized using the same pixel size as the TIR rasters. Then these different raster representations of the cwp were stacked into a multi layer raster and a pixel wise mean calculation was performed. This aggregates the three dimensional raster stack into a two dimensional raster. The pixel wise mean computation produces a pixel score between 0 and 1 which can be interpreted the following way. A pixel value of 1 indicates that this area of the emme basement, was part of a cwp during each of the runs. A pixel value near zero indicates that the area covered by the pixel almost never was part of a cwp in the model outputs. The resulting raster can be understood as a fuzzy membership representation of the CWPs in the emme where each pixel has a probability score of belonging to a cwp. From these fuzzy cwps it can also be read which cwps are variable with respect to the step length (or the buffer pixel => which however has a neglectable influence as seen in the results). CPWs which show a lot of

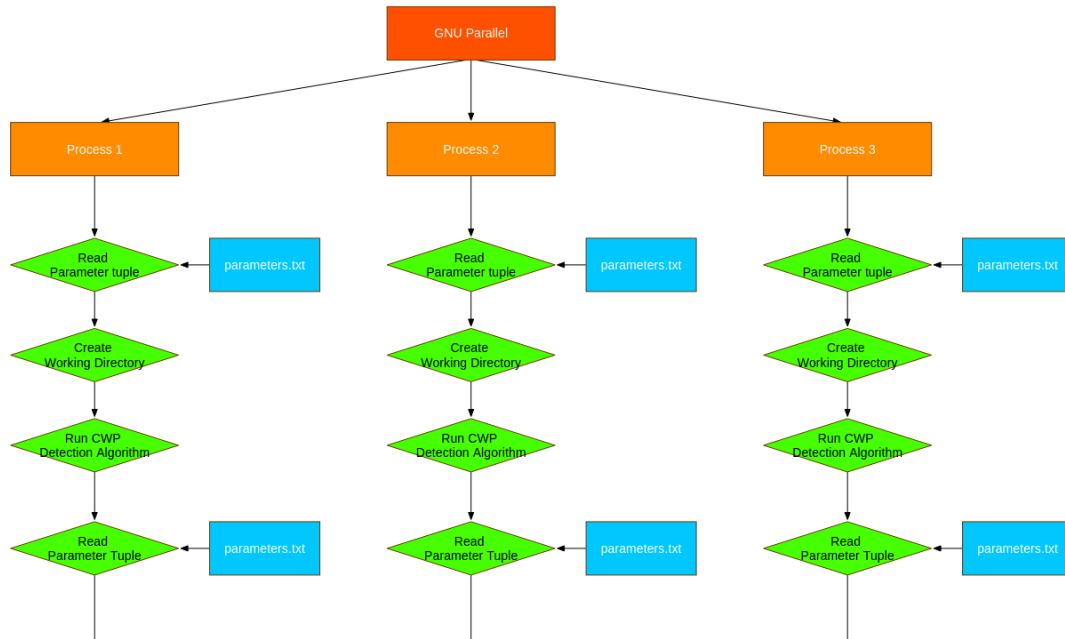


Abb. 2-3: TODO: describe what this plot shows.

differnt shades in their colouring, hold pixels which are generally variable in their identity as cwp pixel or non cwp pixel. These are then cwp that seem to vary with respect to bla bla

Selection and Classificatio of CWP

From this fuzzy cwp raster, cwp pixels which were clearly variable and which mode of creation could clearly be attributed to tributaries or not tributaries were then selected and classified into groups tributary-caused and non tributary-caused. The selection and classification process was not carried out algorithmically due to time constraints but was based on a visual interpretation of the patches. First patches were judged based on their coloring as being variable or non variable. For these patches were generally determined as variable, when their coloring held a significant amount of grey and black pixels. These cwp then underwent classification. For a cwp to be tributary caused, the closeness to a tributary inflow as well as the shape of the cwp were taken into account. Cwp clearly originating in their shape from a tributary inflow and being close to the inflow were determined as tributary caused. To detect the tributaries, an open street map as well as the Swiss image were deployed. The open street map facilitated the possibility to spot tributaries which might be obscured by forest or buildings on the Swiss images, while the Swiss image provided the opportunity to spot tributaries which might not be mapped on the open street map as the open street map is based on volunteered geographic content and therefore can not be assumed as being complete. A total of 24 cwp patches which are variable and can be clearly attributed to being tributary caused or non tributary caused were selected and classified. In the process, their bounding box was extracted and an image was taken. Figure xzy describes the classification process on a concrete example.

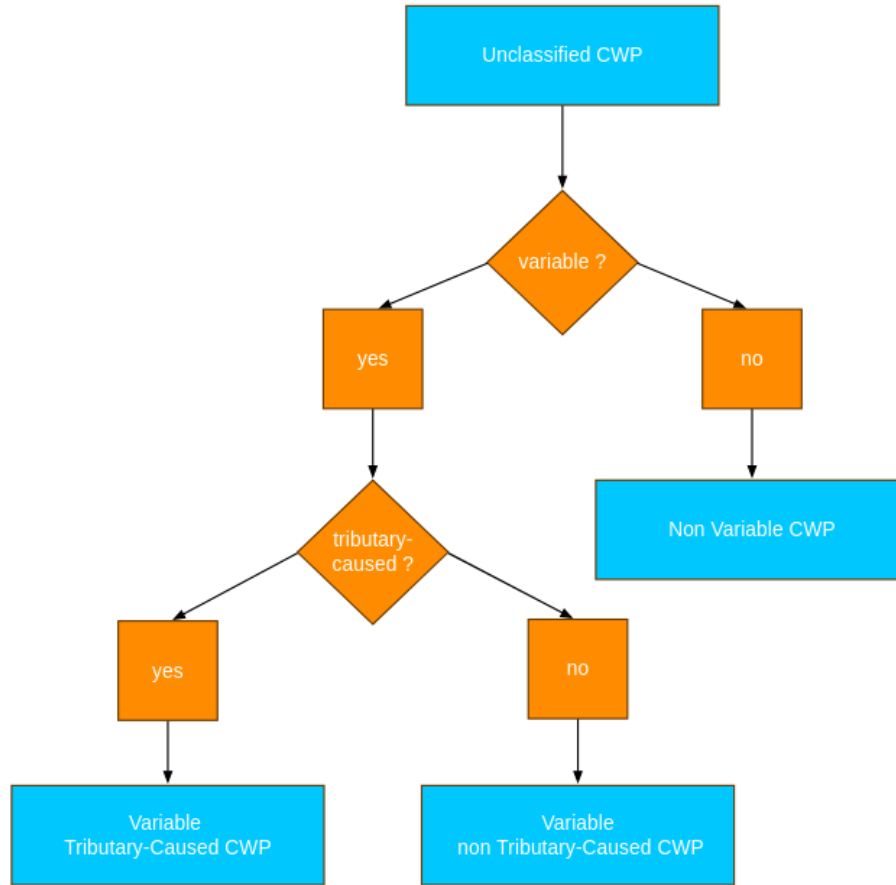


Abb. 2-4: Decision tree illustrating the visual classification process applied to candidate CWP patches: patches are first classified as variable or non-variable based on coloring, and variable patches are further classified as tributary-caused or non-tributary-caused based on their proximity to, and shape relative to, a tributary inflow.



Abb. 2-5: TODO: describe what this overview plot shows.

3.2.2 Computing morris sensitivity indices with respect to CWP area

Setting up second sampling Grid and executing Algorithm

with these cwps classified, a second morris experiment was set up this time also varying the temperature delta parameter next to the buffer px and the step length. A visual depiction of the second morris sampling grid is provided in Figure xzy. A total of 64 trajectories were sampled from this grid creating a total of $64 * (3 + 1) = 256$ sampling tuples. Then the model was again executed in parallel on the different parameter tuples with the remaining parameter held constant, which are provided in table xyz. For the execution orchestration, the same set up as in 3.2.1 was deployed. The resulting cwp polygons were again written to folders uniquely identified by an id, as well as the parameter values.

Computing specially lumped statistics for the selected CWPs

In the next step the extracted bounding boxes locating the area of the selected cwp aswell as the results from this larger morris experiment are brought together. For each bounding box, the CWP/CWPS lying inside this bounding box are extracted out. Next the total area covered by CWP in the bounding box is computed and the area weighted mean temperature delta of all CWPS ins the bounding box are computed. If there is only a singular CWP in the bounding box, then this just corresponds to the temperature delta of this singular CWP. If a bounding box is completely empty for a run with a specific parameter combination, which happens when for this specific run no CWP at all was computed at this location, then the total area is set to zero m2 and the temperature delta is set to zero degrees celsius. The later is a trivial approach, where no CWP is of course equivalent to a CWP with zero surface area. The second one makes sense in the sence that a CWP with no temperature difference to its surrounding per definition can not be a CWP and therefore does not exist. So a temperaure delta of zero for a non exsiting CWP makes logical sense.

The area and temperature delta measure is then finaly paired with the morris index of the parameter tuple and the identifier of the bounding box. This way the measures can be linked back to the parameter combination that produced them and the location of the bounding box. Also the determined class of the cwp is added to this too. Figure 2-6 is providing a graphical overview of this process using a CWP created though the Limpbach inflow into the emme near Utzenstorf BE.

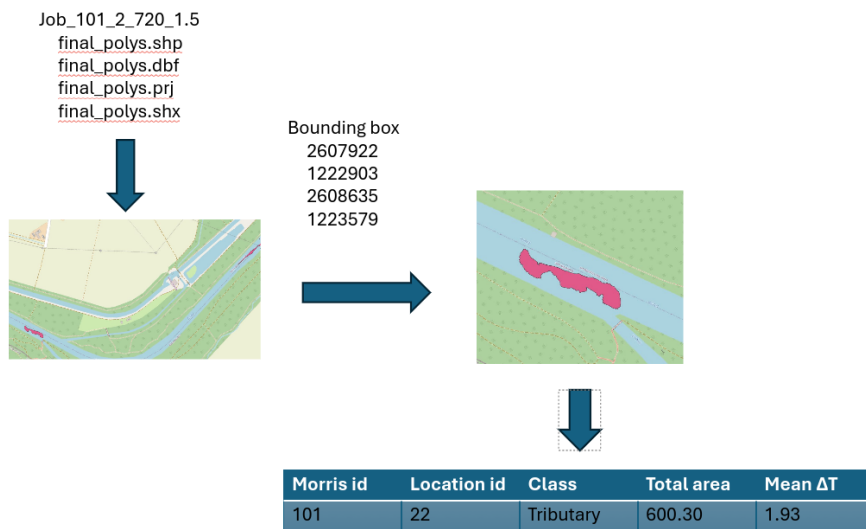


Abb. 2-6: TODO: describe what this overview plot shows. And generally make a better plot

As visible in the table, do the area measures generally differ for the same cwp location due to the different parameters used. Therefore the cwp area is a suited scalar statistic to determine the variability of the cwp patches with respect to the input parameters.

Computing morris sensitivity indices

Lastly, the morris sensitivity indices μ_{star} , μ and σ are computed with respect to the total CWP areas for each bounding box.

To later on make the calculated morris indices comparable to each other normalisation of the patch area was needed. This since without normalisation, large patches would dominate smaller patches just due to their size. So for every location/bounding box Min max scaling into the zero one range was performed. where the largest area value ever produced for this bounding box over all computational runs corresponds to 1 and zero corresponds to the case when the bounding box is completely empty in a specific run.

Computation of the morris indices was then performed with respect to these total area measures by again utilising the sensitivity package. As introduced in the theoretical background does morris produce one ensitivity index (μ , μ .star or σ). for each parameter that was varied in the experiment. So for every location/ bounding box recorded two sensitivity values (μ .star and σ) are produced for every parameter. So a total fo 6 parameters.

these sensitivity parameters where then lastly collected and displayed against each other

Visualisation and statistical modeling

A set of boxplot visualisations displaying μ star and σ with regards to different parameters where produced. the plots differentiate the two groups tributary-caused and non tributary-caused as seperate boxplots.

Additionally poxplots of the μ .star distirbution with regards to all three parameters buffer pixel, step length and temperature delta where produced whithout differentiating between the classes.

A linear model based the mean area of all CWP realisation of a boundinbox and the mean temperatue delta of all cwp realisations of a boundingbox was then fitted to predict the sensitivity value μ .star with respect to the step length parameter. This was done to gain insight into wheter the area of the patch and the temperature difference of the patch relate and therefore might explain the sensitivity to the step lenght parameter.

3.3 Usage of KI

Claude Opus 4.6 was used as an aid to ensure proper formulation, spelling and grammatical correctness. Further it was used to derive the LaTeX for Images and simulation protocol table.

Chapter 4

Results

4.1 Comparison of outputs

The three comparison runs achieved outputs that are in general agreement of the cwp location and shape but are not on a pixel perfect basis. The jaccard similarities for the three runs are 0.936, 0.956 and 0.976 which shows generally good agreement between the methods.

Figure 1-1 shows the cwps generated during trial run 3. In red the polygons of the speed optimized algorithm, in red the polygons of the original algorithm. It can be seen that for the displayed cwps, the two algorithms are in general agreement, however do not fully agree on the pixel level. Figure xzz shows a more pronounced version of this disagreement.



Abb. 1-1: CWPs generated during trial run 3. Polygons of the speed-optimized algorithm are shown in red, and polygons of the original algorithm are shown in blue. The two algorithms are in general agreement, but do not fully agree at the pixel level.

Figure 1-2 shows an example of more pronounced disagreement between the two functions for trial run 1, figure 1-3 shows an example set of cwps with perfect agreement between

the two methods



Abb. 1-2: TODO



Abb. 1-3: TODO

4.2 Speed comparison in execution

Figure 2-4 shows the execution speed of the original as well as the speed optimized cwp function. On the y-axis the execution speed in hours is plotted, on the x-axis the step-length parameter is plotted in meters. Execution speed is a constant half hour for the speed optimized version where the original function ranges between 2 and 1.25 hours for the three runs with higher execution times for shorter step lengths.

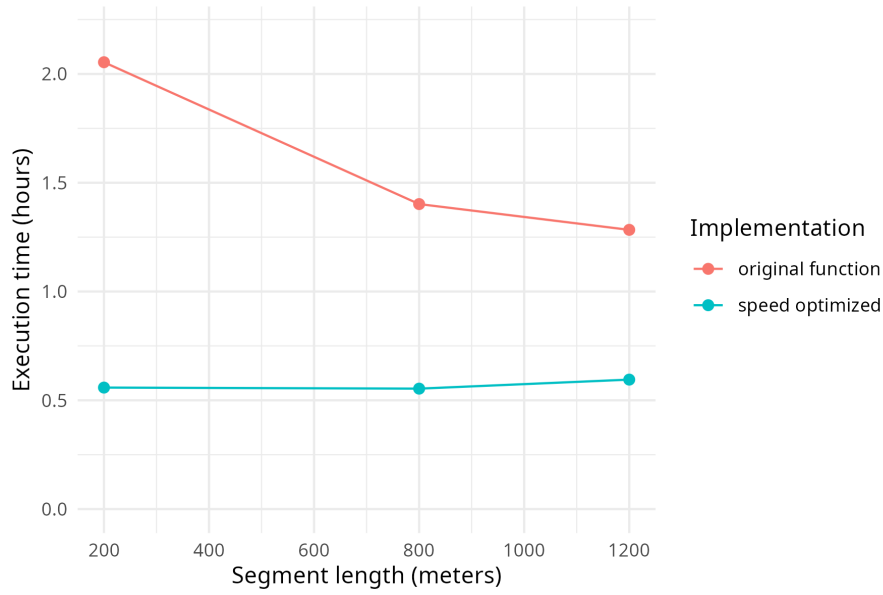


Abb. 2-4: Execution time of the original (v1) and optimized (v2) implementations of `detect_cwp_single` as a function of segment length. The optimized implementation shows execution times that are both shorter and largely constant across segment lengths, whereas the original implementation scales disproportionately with decreasing segment length (i.e., increasing number of segments).

4.3 Sensitivity scores for step length parameter

Figure 3-5 shows morris sensitivity indices sigma and mustar with regards to the step length parameter. It can be seen that all non tributary caues CWP have a sensitivity index below 0.2. The spread in mu star for the sensitivity caused cwp is larger with some cwp featuring a similar sensitivity as the non tributary caused cwp and some featuing a substantially larger sensitivity value (0.3 and 0.6). The mean of both mu.star boxplots are very similar for both classes.

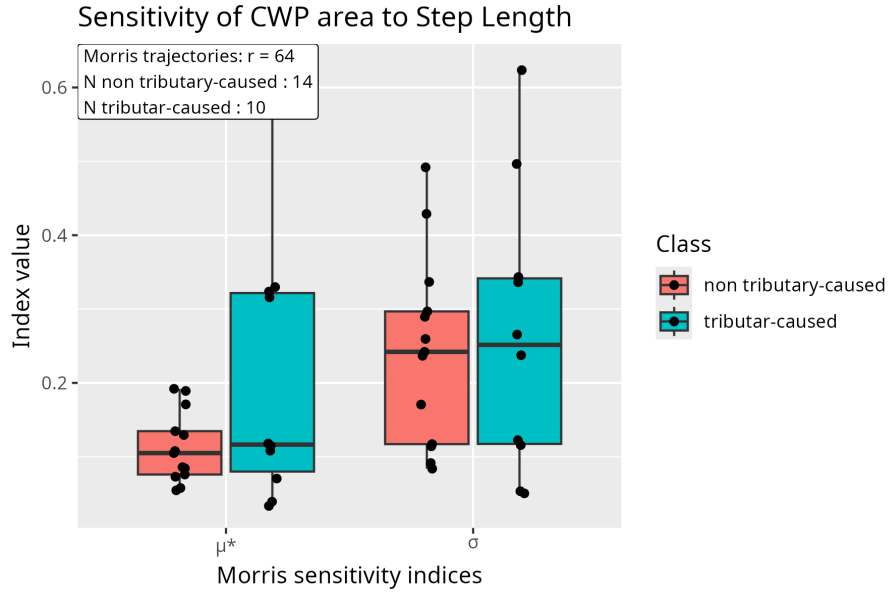


Abb. 3-5: Morris sensitivity indices (μ^* and σ) for the `step_length` parameter with respect to cold water patch area, stratified by patch class (red: tributary-caused; blue: non-tributary-caused). Boxplots show the distribution across bounding boxes, with sample sizes (n) given below each box.

4.4 Mu star for all parameters over all CWP

Figure 4-6 displays the mu.star values of all cwp with respect to the three different parameters varied during the morris experiment. The buffer pixel count continuously scores very low mu.star values while the step length and the temperature delta deviate from zero. Especially the temperature delta seems to highly influence the cwp with respect to their area.

4.5 Mu.star and size for every cwp

Figure 5-7 shows the 24 selected cpw along 3 different dimensions. On the x axis, the averaged cwp area over all morris runs is depicted, on the y axis the sensitivity against the step length parameter is depicted. The color is used to indicate the average temperature delta to the surrounding environment averaged over all morris runs is depicted. The shape is indicating the class of the cwp.

Tab. 5-1: Regression results for μ^* predicted by mean area and mean ΔT

Predictor	Estimate	SE	t	p
(Intercept)	9.024×10^{-2}	3.094×10^{-2}	2.92	0.009
Mean area	1.578×10^{-4}	2.839×10^{-5}	5.56	< 0.001
Mean ΔT	-1.041×10^{-2}	1.739×10^{-2}	-0.60	0.556

Note. $R^2 = 0.611$, Adjusted $R^2 = 0.572$, $F(2, 20) = 15.73$, $p < 0.001$. Residual standard error = 0.088 on 20 degrees of freedom (one observation deleted due to missingness).

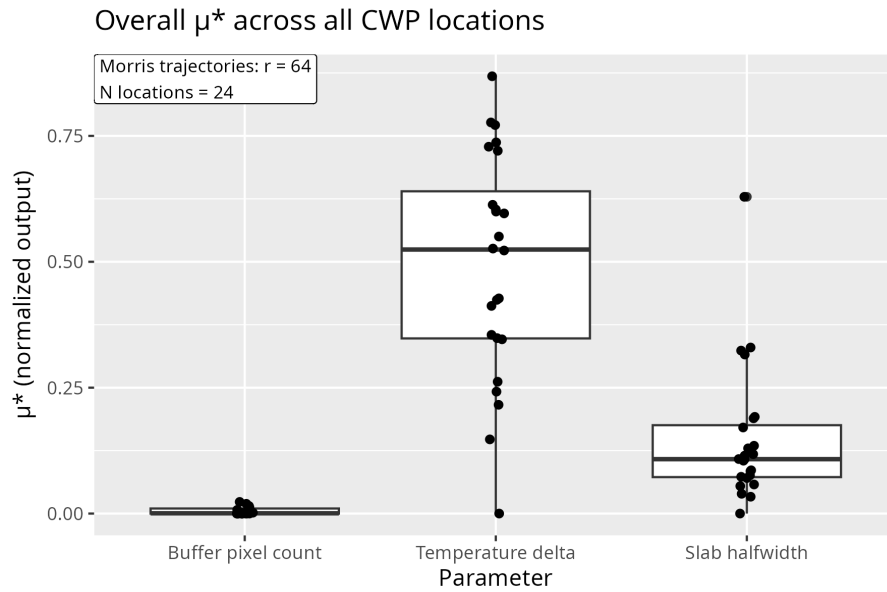


Abb. 4-6: Overall μ^* sensitivity indices for `buffer_px`, `delta_T`, and `step_length` across all classified cold water patches ($N = 24$ locations, $r = 64$ trajectories).

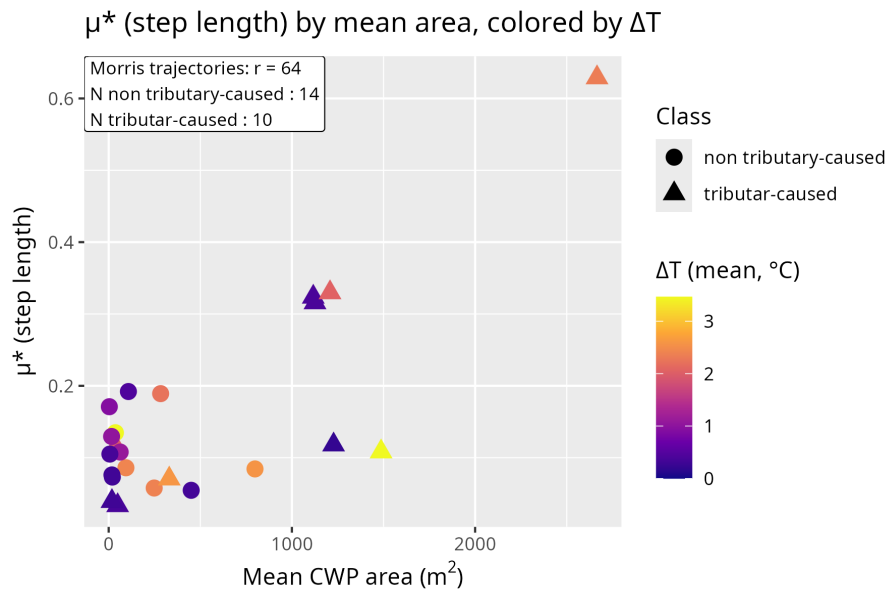


Abb. 5-7: μ^* (step length) versus mean CWP area for each annotated location, colored by patch class.

Chapter 5

Discussion

5.1 Difference in cwp polygons between old and new function

the cwp patches produced by the old and the speed optimized function are generally comparable in their position and extent, but are not completely equivalent on the pixel level. Generally the obtained jackard similarities are high. This is not optimal, however with the respect to the research questions it can be tolerated. The research question generally asks for the effect of the step length parameter on tributary caused or non tributary caused cwp. It is therefore not completely bound to the original algorithm, but more to the concept of cwp. As long as the optimized algorithm reflects the physical reality and flags reasonable cwps, then this is ok. And it does flag ecologically reasonable cwp as it produces very similar cwp as the original cwp algorithm which can be assumed as being ecologically plausible. The disagreement is mostly on the pixel level, this might be due to rounding. Especially the different behaviour of `terra::extract` and `exactextractr::extract_extract` could cause this issue because their difference in including/excluding partially covered pixels might change the reference temperatures enough, to cause a difference which is then visible at the pixel level. Generally it could also be another rounding error. It has to be something minor, as only a small set of pixels at the border of the patches usually is affected and generally also a range of cwp exist, that are not affected by the problem at all. So rounding issue or something is plausible

5.2 Speed comparison

The optimized cwp algorithm shows reasonably faster execution speeds for all tree runs. Also what is of interest is, that the execution speed of the speed optimized cwp algorithm is constant, while the execution speed of the original function grows with smaller step lengths. This would make mechanistic sense. The speed optimized function does not deploy any looping. Instead it performs a number of operations on every pixel that is independent on the number of steps being used. The original function uses R looping to process the data. There might be some constant overhead connected with each loop and bla bla but generally it is hard to say. It was a good decision to optimize the function as the range of the morris experiment ranges from 400 to 1200 and the optimized algorithm is faster in the complete range with being significantly faster for 400 meters.

5.3 Answering research question 1

The boxplots show that there is a tendency that the tributary caused cwps are more sensitive to the step length parameter than the non tributary-caused. The mean of the boxplot is slightly higher and the third quantile as well as the maximal value is significantly higher. Upon closer inspection however it is visible, that some tributary caused cwp behave very much alike the non tributary cwps in their sensitivity while a second group seems to be vastly more sensitive. It is therefore questionable, whether the pure identity of being caused by a tributary is telling the whole story when it comes to governing sensitivity to step length or whether there are further effects at play within the tributary-caused group that ultimately govern the sensitivity with respect to the buffer length. In other words, being caused by a tributary could be an important prerequisite of being sensitive to the step length parameter, however it is not the complete and ultimate cause of the sensitivity itself.

And this interpretation can find support in the scatter plot. Given by the coloring reflecting the temperature difference of the cwp to the surrounding stream, it seems that both high ΔT cwp as well as low ΔT cwp seem to be insensitive to the step length parameter. Additionally there are also certain cwp with intermediate sensitivity to the step length parameter (μ star of 0.3) which have both high as well as smaller temperature deltas. So by the general mixing of colors on the plot, it does not seem like the temperature delta has not much of an influence on whether a cwp is sensitive to the step length parameter. The data however suggests that higher cwp areas drive the sensitivity to the step length parameter. To be more specific the data seems to suggest a linear relationship between mean cwp area and the step length sensitivity. This visual interpretation is also supported by the linear model fitted to the data. The slope derived for the mean_area is highly significant while the slope derived for the mean temperature delta is not significant with a p value of 0.55 indicating that this value is likely to be observed when no correlation between the temperature delta and the sensitivity to step length is assumed. An important caveat to take into account is the small sample size of $n=24$. While the highly significant parameter for the cwp area on the one hand supports, that the linear relation is strong and already detectable with a small number of parameters, it is still also important to understand, that the chance of accidentally sampling a linear relation when a small number of points are sampled is higher, than when a large number of samples are drawn.

But generally the data would support the following claim: CWP caused by tributaries are not automatically highly sensitive to the step length parameter. What really counts is the size of the CWP. Being tributary caused seems merely to be a driver to being a large CWP, since a tributary might introduce large amounts of colder water into the river which in turn could be a prerequisite for large CWPs. So being tributary caused can be a good driver for a cwp to be large. And large cwps are mainly long, this because the emme is a river and rivers are usually much longer than they are broad so in order to have a large area, a patch generally needs to be long while gaining large area from becoming more broad is limited. So large cwps have a tendency to be long and if a cwp is long then the chance that the border between two slaps falls within the patch is larger. and this is then the driver in the susceptibility to the step length parameter. falling into different slabs as this yields different reference temperatures which are used for comparing.

From the data it can also not be said whether a large non tributary-caused cwp would be equally susceptible to the step length as equally large tributary-caused cwp. This because there is no non tributary caused cwp larger than 1000 m² in the data present.

Interestingly the temperature difference and the area of the cwp seem not to be related. This is rather interesting since I automatically assumed that a cwp can potentially grow larger if a tributary provides cold water. Maybe in order to determine that the discharge of the tributary would be interesting to know during data acquisition. It could be that the optimal condition for a large cwp and therefore a sensitive cwp is a tributary with large discharge and a low temperature.

But the data would generally support the following answer to research question 1:

5.4 Overall Sensitivity to the different Parameters

It is clearly visible that the buffer pixel number seems to be non influential with respect to the cwp area. This finding is also supported by Tonella et al., which found the same result in their sensitivity analysis. Also there seems to be little potential interaction with other parameters. For this reason, factor fixing can be deployed and the buffer pixel count can be set to a specific value.

The Temperature delta set for the pixel flagging generally seems to be the most influential parameter with regards to the cwp area. This finding is also consistent with Tonella et al.

The temperature delta seems generally be the most influential

Chapter 6

Optimisation of the CWP algorithm based on the morris sensitivity analysis

6.1 Deploying insight from the morris screening

The results of the morris sensitivity analysis have shown that the buffer pixel seems to be largely non influential on the area of cwp. Therefore the strategy of factor fixing can be performed, and the buffer pixel parameter can be internalized and be set to a constant value which in this case was choosen to be 3 which lies within the range of the morris screening grid.

Further it is evidential, that the step length seems to have a reasonable influence on the computed patch area both for the tributary caused and non-tributary caused cwp. It is therefore of interest to remove the dependency of the algorithm on this parameter aswell.

Laslty the temperature delta parameter seems to have the strongest effect of all tested parameters. This parameter however can not be simply reduced away, as it has to be set by the user. This because in literature different temperature deltas for the definition of a cwp exist and therefore a user needs to be able to pick the one that is corresponding with his or heres definition.

So summing up the new algorithm can remove the buffer pixel parameter by factor fixing and has to provide a new approach of detecting ecologically reasonable cwp without relying on a step length parameter.

6.2 Operationalisation of Surrounding Stream temperature

The later was achieved by rethinking the operationalsiation of the idea of surrounding stream temperature as given by the ecological definition. Instead of relying on the mean temperature of fixed slabs, the new algorithm deploys a inverse distance weighting scheme to compute a reference temperature for each cell by deploying a inverse distance weighting scheme. This guarantees that the reference temperature for pixel flagging is always a mutliperspective decission taking surrounding temerature values of different locations into account to provide a surrounding temperature. This strategy as the added benefit that

the deployed reference temperature is a spatially continuous phenomenon unlike the older purely slab based approach. Therefore sharp cutoffs of cwps at slab boundaries can no longer occur unlike in the old algorithm

6.3 Implementation Detail

generally the implementation to a large degree follows the implementation of the time optimized cwp algorithm only differing from said function in the computation of the reference temperature step.

the following pseudo code provides a detailed overview of this new reference temperature computation

Step 4. is thereby written in C++ and imported into R using the package Rcpp. This, as it was identified as a computationally intensive step which is best implemented in an compiled language. To further increase the execution speed of this step, parallelisation on the thread level was deployed, as the inverse distance computation for very pixel is an embarisingly parallel computation. For the parallelisation stepup, the OpenMP library was deployed.

6.4 Reduction of parameters

as mentioned, the buffer px parameter was fixed. Due to the distributed reference points and the inverse distance weighting, no zone slab length is needed anymore. Additionally, since the extent of the reference raster is allways mached to the extend of nodata pixels in the TIR raster due to 5.1, a slab half width also does not have to be set. This however also means, that the algorithm can only be applied to exact water masks of rivers and the shore needs to be cut out. The remaining parameters are the paths to the raster and the centerline, the minimal patch area, the rounding factor, the temperature delta, wheter diagonal pixels should be counted as neighbours in rasterisation. Additionally a new parameter was introduced, which lets a user set the number of cores used for internal parallelisation.

6.5 Quellenverzeichnis

- E. Dürig. Raster deepdive, 2026. URL <https://stds-handin.github.io/Raster-Deepdive/>. Coursework submission; not peer-reviewed, cited as supporting empirical observation only.
- GDAL/OGR contributors. GTiff – GeoTIFF file format, 2026. URL <https://gdal.org/en/stable/drivers/raster/gtiff.html>.
- R. J. Hijmans, A. Brown, and M. Barbosa. extract: Extract values from a SpatRaster, 2024. URL <https://rspatial.github.io/terra/reference/extract.html>. terra package documentation.
- ISciences, LLC. exactextract, 2026. URL <https://isciences.github.io/exactextract/index.html>.
- A. Ligmann-Zielinska. Spatially-explicit sensitivity analysis of an agent-based model of land use change. *International Journal of Geographical Information Science*, 27(9):1764–1781, Sept. 2013. ISSN 1365-8816, 1362-3087. doi: 10.1080/13658816.2013.782613. URL <http://www.tandfonline.com/doi/abs/10.1080/13658816.2013.782613>.
- R. Lovelace, J. Nowosad, and J. Muenchow. *Geocomputation with R*. CRC Press, 2019. ISBN 9780203730058.
- T. Mieno. R as GIS for economists: Extraction speed comparison, 2022. URL <https://tmieno2.github.io/R-as-GIS-for-Economists/extract-speed.html>.
- A. Saltelli, editor. *Sensitivity analysis in practice: a guide to assessing scientific models*. Wiley, Hoboken, NJ, 2004. ISBN 978-0-470-87093-8.
- A. Saltelli and P. Annoni. How to avoid a perfunctory sensitivity analysis. *Environmental Modelling & Software*, 25(12):1508–1517, Dec. 2010. ISSN 13648152. doi: 10.1016/j.envsoft.2010.04.012. URL <https://linkinghub.elsevier.com/retrieve/pii/S1364815210001180>.
- A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, and S. Tarantola. *Global Sensitivity Analysis. The Primer*. Wiley, 1 edition, Dec. 2007. ISBN 978-0-470-05997-5 978-0-470-72518-4. doi: 10.1002/9780470725184. URL <https://onlinelibrary.wiley.com/doi/book/10.1002/9780470725184>.
- O. Tange. GNU Parallel 20240222. Zenodo, Software, Feb. 2024. URL <https://doi.org/10.5281/zenodo.10719803>.
- D. Tonolla and M. Antonetti. Tonolla_etal_2021_schlussbericht_zhaw_thermoemme. Technical report, ZHAW, Wädenswil, June 2021.
- D. Tonolla and M. Antonetti. Thermische-Kartographie zur Detektion und Charakterisierung von Kaltwasserbereichen an der Emme und drei ausgewählten Zuflüssen (Grüene, Ilfis, Urtene). Technical report, ZHAW, Wädenswil, Feb. 2026.